

ShopFlow AWS Project

DynamoDB Design

NoSQL Tables - Session Store - Idempotency Keys - TTL

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Field	Value
Version	1.0
Date	June 2026
Region	ap-south-1 (Mumbai)
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Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

Amazon DynamoDB provides serverless NoSQL storage for ShopFlow's session management and idempotency tracking. DynamoDB requires no capacity planning -- it scales automatically. ShopFlow uses On-Demand billing mode (pay per request), making cost proportional to usage.

2. Tables

Table Name	Primary Key	GSI	TTL Attribute	Purpose
shopflow-sessions-dev	tokenHash (PK)	userId-index	expiresAt	JWT session store -- validate token on each request
shopflow-idempotency-dev	idempotencyKey (PK)	None	expiresAt	Prevent duplicate order processing

3. Session Table -- Item Schema

Attribute	Type	Description
tokenHash	String (PK)	SHA-256 hash of the JWT token -- unique per session
userId	String	User UUID -- queried via GSI for 'logout all devices'
email	String	User email address
expiresAt	Number	Unix timestamp -- DynamoDB auto-deletes when this passes
createdAt	String	ISO 8601 creation timestamp

Global Secondary Index -- userId-index

The GSI allows querying all sessions for a specific user. Use case: 'logout from all devices' -- query by userId, delete all returned sessions.

4. Idempotency Table -- Item Schema

Attribute	Type	Description
idempotencyKey	String (PK)	Client-generated UUID sent in request header
status	String	COMPLETED IN_PROGRESS
responseBody	String	Cached JSON response -- returned on duplicate request
expiresAt	Number	Unix timestamp -- auto-deleted after 24 hours

5. Idempotency Flow

Request	DynamoDB Action	Outcome
First request	PUT item with status=IN_PROGRESS	Order processed normally

Request	DynamoDB Action	Outcome
Same key, still running	Item exists with IN_PROGRESS	Return 409 Conflict
Same key, already done	Item exists with status=COMPLETED	Return cached response -- no duplicate order
Key expired (>24h)	Item deleted by TTL	Treat as new request

6. TTL (Time To Live)

TTL is free. DynamoDB reads the `expiresAt` attribute (Unix timestamp). When the current time exceeds `expiresAt`, the item is marked expired and deleted within 48 hours. Expired items do not appear in read results immediately after expiry.